Free and Fair? A Systematic Study of Reliability and Bias in Open-Source LLM Judges across Families and Scales

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Abstract

Free-tier LLM APIs make large-language-model judges accessible to any budget, yet the reliability of these freely usable referees is largely uncharacterised. We audit ten open-source judges from four model families (Qwen, DeepSeek, GLM, Kimi) plus two proprietary free-tier references under one protocol: 400 answer pairs spanning mathematics, factual QA, creative writing, and abstract explanation, each judged in both presentation orders, with a test–retest subset and ground-truth-controlled family-preference probes, all at zero monetary cost. Four findings stand out. First, severity collapse: DeepSeek flash and flagship judges abstain with ties on 50–59% of pairs, dragging their ground-truth accuracy as low as 0.32. Second, instruction backfire: a standard anti-verbosity sentence inverts the usual verbosity bias, producing significant preference for the *shorter* answer (-0.46 log-odds per 1,000 characters). Third, family-specific kinship: Qwen judges significantly avoid same-family answers while GLM and Kimi reward them even when the reference answer proves them wrong. Fourth, panel gains without debiasing: majority voting over three judges raises ground-truth accuracy from 0.58 to 0.85 but leaves the shared position-bias component essentially unchanged, so

order swapping remains necessary. We distill the results into six deployment guidelines and release the complete artifact—questions, prompts, 9,453 verdicts, and analysis code—so the audit can be re-run as free-tier catalogues evolve.
Keywords: LLM-as-a-Judge, evaluation reliability, position bias, self-preference bias, open-source models

1. Introduction

Large language models (LLMs) are increasingly asked not only to produce answers but also to grade them. The LLM-as-a-Judge paradigm, in which a model compares or scores candidate responses in place of human annotators, has become a workhorse of modern language technology. It powers public leaderboards built on pairwise comparison [1], substitutes for human preference labels in instruction tuning pipelines, and provides cheap regression tests for retrieval-augmented systems, agents, and text generation services [2, 3]. As the scale of evaluation grows far beyond what human rating budgets can cover, the quality of machine judgement itself has become a first-order scientific question.

The dominant practice rests on a small set of closed, frontier models such as GPT-4 and Claude. This reliance carries well-known costs: per-token fees that grow linearly with evaluation volume, restricted data governance, and reproducibility risks when a provider silently updates a checkpoint. Open models offer an appealing alternative. Their weights are public or freely hosted, and several providers expose capable open families through zero-cost API tiers, which makes them attractive to academic groups and small

engineering teams. Whether these freely accessible models can actually serve as dependable judges, however, remains poorly characterized. Existing bias studies concentrate on closed judges from a handful of providers [1, 4, 5], and the largest recent audit of 21 judges covers nine commercial providers while leaving open-weight families at the periphery. Openness does not imply neutrality: a judge trained by the same organization that produced some of the candidates may share stylistic priors, tokenization habits, or even training data with those candidates, which creates a structural risk of family favoritism that earlier self-recognition studies only partially capture [6].

This paper presents a systematic anatomy of reliability and bias for freely accessible LLM judges. We assemble ten open-source judges drawn from four model families (Qwen, DeepSeek, GLM, and Kimi), spanning multiple capability tiers within each family, and complement them with two proprietary free-tier judges (Agnes-2.5-Flash and Agnes-3.0-Flash) as a reference group, evaluating all of them under one unified protocol. Five generators, one per family plus a second Qwen tier, answer 200 questions from four domains: grade-school mathematics with verifiable ground truth [7], adversarial factual QA with reference answers [8], open-ended creative writing, and abstract explanation. From these answers we construct 400 stratified answer pairs, and every judge scores every pair in both presentation orders, yielding 9,453 usable verdicts, with an additional test-retest subset of 640 repeated judgements on the open judges to probe stochastic stability.

The design isolates four bias families with matched measurements for every judge. Position bias is measured directly by comparing verdicts on swapped presentation orders, length bias by the conditional win rate of the

longer answer, accuracy on ground-truth-clear pairs in the two verifiable domains, and family preference by a quality-controlled protocol that tracks how often a judge rewards a same-family answer when the reference answer proves that answer wrong. Inter-judge structure is summarized by pairwise Cohen's kappa [9], and all headline estimates carry question-level cluster bootstrap confidence intervals.

Four findings structure the results. *Severity collapse*: the DeepSeek flash and flagship judges return ties on 50–59% of pairs, silencing their discrimination and pushing ground-truth accuracy below chance, a failure mode invisible to compliance checks. *Instruction backfire*: the standard anti-verbosity rubric sentence inverts the usual verbosity bias—every Qwen, GLM, and Kimi judge tilts against the longer answer, significantly so in the pooled regression. *Lineage-specific kinship*: Qwen judges avoid same-family answers (37% kin win rate) while GLM-5.3 and Kimi-K3 reward theirs even when the reference proves them wrong (27–33% kin-wrong pick rate). *Panel gains without debiasing*: majority voting over three free judges lifts accuracy from 0.58 to 0.85 while leaving the shared position-bias component essentially unchanged, showing that panels are accuracy instruments, not debiasing ones.

Beyond measurement, we translate the findings into deployment guidance for practitioners who cannot afford commercial judge APIs. We quantify how majority panels of three or five open judges trade compute for accuracy and bias robustness, revisit the order-swapping protocol as a cheap debiasing device, and identify which free-tier models are safe to use as referees for which kinds of tasks. All prompts, question sets, raw verdicts, and analysis code are released with the paper to support reuse.

Our contributions are fourfold. First, we provide the first systematic, multi-family, multi-tier study of judging reliability for freely accessible LLM judges, covering four bias families under one protocol and contrasting open-source judges with proprietary free-tier references. Second, we introduce a quality-controlled kin-preference measurement that separates family favoritism from genuine quality differences. Third, we give within-family scale-gradient evidence that links judge behaviour to model tier rather than to family identity alone. Fourth, we distill the results into concrete, zero-cost judging guidelines and release the complete experimental artifact.

2. Related Work

2.1. LLM-as-a-Judge

Using LLMs to evaluate text quality was popularized by G-Eval, which framed NLG scoring as a probability-weighted form-filling task with chain-of-thought [2], and by MT-Bench, which established pairwise comparison with a reference-guided rubric as the default protocol and reported the first catalogue of judge failure modes [1]. Subsequent surveys systematized the design space of judges, prompts, and aggregation schemes [3]. A complementary line of work trains open models specifically for evaluation: the Prometheus family fine-tunes open LLMs on human feedback to produce reproducible graders whose agreement with humans rivals that of general-purpose judges [10]. Our study differs in orientation: rather than proposing a new judge or benchmark, we audit the judging behaviour of off-the-shelf open models that practitioners can already call for free, since in practice most teams use unmodified chat models as referees rather than dedicated evaluator check-

points.

2.2. Judgement Biases

Three bias families dominate the literature. Position bias, the tendency to favour whichever answer appears first (or last), was demonstrated early for GPT-4-class judges and motivates order-swapping or consistency-based mitigations [1, 11, 12]. Verbosity bias, the correlation between answer length and preference, was documented in RLHF preference labels and in arena-style evaluation, and motivated length-controlled win rates [13, 14]. Self-preference is the subtler tendency of a judge to favour its own generations: evaluators have been shown to recognize their own text and to reward it, even when the text carries a hidden watermark that the judge cannot consciously detect [6]. Comparative studies of human and machine judges further show that some biases, such as self-preference and conformity to anchor answers, are shared by humans and LLMs, while others, such as opinionatedness, differ sharply [4]. We generalize the self-preference lens from the individual model to the model family: because several providers ship many tiers of the same base model, a practical question is whether a judge favours any member of its own lineage, which we measure with ground-truth controls.

2.3. Reliability, Validity, and Judge Panels

A growing body of work asks whether judges agree at all. Cross-judge agreement with humans is often modest and task-dependent [15, 16]. A recent large-scale audit of 21 judges from nine providers reports that exact-match agreement overstates discriminative ability relative to Cohen's kappa, that judge rankings are unstable across benchmarks, and that high test-retest

reliability can coexist with severe position bias [5]. Infrastructure work such as JudgeArena unifies judge benchmarks and ships tuned configurations for open judges [17]. On the mitigation side, panels of diverse judges have been proposed to average away idiosyncratic errors [18], though correlated errors within panels can undermine the diversity argument. Our contribution to this line is a family-and-tier-resolved view of agreement structure among free judges, plus a panel analysis that respects family boundaries, so that panel gains are not confounded by kinship.

3. Experimental Setup

3.1. Model Roster

Table 1 lists the twelve judges and five generators. The ten open-source judges are served through a single OpenAI-compatible endpoint (Alibaba Cloud Model Studio free tier) under identical API settings, which removes serving differences as a confound. The open roster spans four families: five Qwen checkpoints covering two flash generations, a 27B mid tier, a max tier, and a 2.4T-parameter Mixture-of-Experts flagship [19]; three DeepSeek checkpoints covering two flash tiers and a pro tier [20]; the GLM-5.3 flagship [21]; and Kimi-K3 [22]. The same five generators produce candidate answers, one flagship per family plus a second, deliberately smaller Qwen tier so that within-family quality variance exists for the kinship analysis. Two judges (Qwen3.8-2.4T and GLM-5.3) are served with mandatory reasoning, a platform restriction of the free tier; we account for this when interpreting their behaviour, and treat *thinking mode* as an observational variable rather than a confound to be hidden.

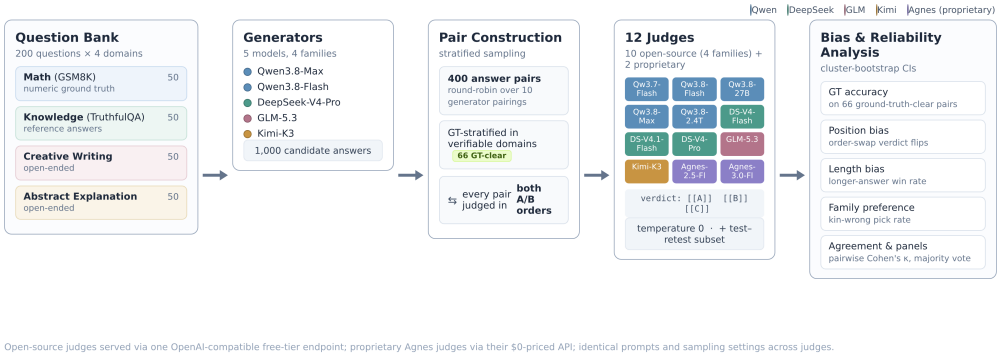


Figure 1: Overview of the evaluation pipeline. Two hundred questions from four domains are answered by five generators from four model families; stratified pair construction yields 400 answer pairs including 66 ground-truth-clear pairs, and twelve judges—ten open-source and two proprietary—evaluate every pair in both presentation orders, enabling direct measurement of position bias, length bias, family preference, accuracy, and inter-judge agreement.

Beyond the open roster, we add two proprietary judges, Agnes-2.5-Flash and Agnes-3.0-Flash, accessed through their public API on the provider's zero-priced promotional tier. These models give a proprietary reference point against which the open judges can be read; they see the identical judging prompt, but their serving platform injects undisclosed system-level instructions ahead of it (observed as roughly 300–600 extra prompt tokens per call), so their absolute metrics are indicative rather than strictly comparable, a caveat we revisit in Section 6.

3.2. Task Domains and Questions

We evaluate on 200 questions, 50 per domain, chosen to combine verifiable and open-ended tasks. Mathematics uses grade-school word problems

from the GSM8K test split, each with a certified numeric answer [7]. Knowledge uses TruthfulQA questions, which probe common misconceptions and carry reference answers [8]; this domain is deliberately adversarial so that generators exhibit meaningful quality differences. Creative writing consists of 50 fixed prompts covering stories, poems, letters, and other open-ended formats, authored for this study and released with the code. Abstract explanation takes 50 recent computer-science abstracts retrieved from arXiv and asks for a plain-language, three-sentence explanation. The two verifiable domains enable objective accuracy measurement, while the two open-ended domains measure agreement where no ground truth exists, mirroring the conditions under which judges are most often deployed in practice.

3.3. Correctness Labelling

Answer correctness in the verifiable domains is determined automatically in two steps. For mathematics, a regular-expression pass extracts the final numeric answer from each response (falling back to the last numeral in the text) and compares it with the certified GSM8K answer numerically after normalization, so that surface forms such as \$162, \$162.00, and 162 are treated as equal. For knowledge, each candidate answer is graded by an independent free-tier labelling model (Qwen3.7-Flash, a checkpoint that is not part of the judge panel, avoiding circularity) that sees the question, the TruthfulQA reference answer, and the candidate answer, and returns a binary CORRECT/INCORRECT verdict with a one-sentence rationale at temperature zero. We validated the labeller against a keyword-matching baseline on manually inspected samples and found substantially cleaner separation, particularly for hedged or partially correct answers; the labeller's verdicts

flipped 29% of knowledge labels relative to the keyword baseline and 8% of mathematics labels relative to a strict string-equality baseline, and manual spot checks confirmed the flips were almost always improvements.

3.4. Pair Construction

From the five generators' answers we form answer pairs by stratified sampling over the ten possible generator pairings. In the two verifiable domains we draw three pairs per question and preferentially pair a wrong answer against a correct one whenever the generators disagree under the correctness screen, which maximizes pairs with an objective quality difference; in the two open-ended domains we draw one pair per question in round-robin rotation. This produces 400 pairs in total, with every generator pairing occurring between 34 and 47 times. The presentation order of each pair is randomized once, and the judging protocol then evaluates every pair in both orders, which turns each pair into its own position-bias probe. In the verifiable domains, a pair is ground-truth-clear when exactly one side is labelled correct by the criteria of Section 3.3; 25% of mathematics pairs and 19% of knowledge pairs are ground-truth-clear (66 pairs in total); and only these pairs enter accuracy and kin-preference analyses. Pair membership is fixed and released with the artifact, and all reported ground-truth quantities use the final labels.

3.5. Judging Protocol

Judges receive an MT-Bench-style rubric: an impartial-evaluator system message, the question, optionally the reference answer (for the verifiable domains only), and the two candidate answers labelled Assistant A and Assistant B. The prompt instructs the judge to weigh correctness, com-

pleteness, clarity, and instruction following, explicitly warns against rewarding length, and requires a one- or two-sentence justification followed by a machine-parseable verdict token: [[A]], [[B]], or [[C]] for a tie. Every judge evaluates every pair in both presentation orders at temperature zero, so any order-induced verdict change is attributable to position rather than sampling noise. Coverage is bounded by each model's free quota: two open judges exhausted theirs near the end of the main pass, and exact per-judge coverage is reported in Section 4. A stratified subset of 40 pairs (ten per domain) is judged twice more in the canonical order to measure test-retest stability. Verdicts are parsed by a strict bracket pattern with one reminder retry; the parse failure rate itself is reported as a compliance metric. To keep the judge input distribution uniform across domains, answers are capped at 900 characters and questions at 1,500 characters with an explicit truncation marker, a setting applied identically to every judge.

3.6. Metrics

For each judge we compute the following quantities, all with 95% question-level cluster bootstrap confidence intervals over 2,000 resamples. Parse rate is the fraction of judgements that yield a valid verdict token. Tie rate is the fraction of canonical-order verdicts equal to [[C]]. Accuracy is computed on ground-truth-clear pairs as the fraction of verdicts that select the objectively correct side. Position bias contrasts the rate at which the first-presented answer wins among non-tie verdicts against the 0.5 null; the flip rate is the fraction of pairs whose canonically-mapped verdict differs between the two presentation orders. Length bias is the rate at which the longer answer wins among non-tie verdicts minus 0.5. Family preference is measured on kin

pairs, where exactly one side comes from the judge's own family: the headline statistic is the kin-wrong pick rate, the fraction of ground-truth-clear kin pairs in which the judge selects its own family's answer although the reference answer proves it wrong. Inter-judge agreement uses pairwise Cohen's kappa on the three-class verdicts of canonical-order judgements [9]. A pooled logistic regression with judge fixed effects estimates the simultaneous effects of length difference, kinship, and an answer-quality difference indicator (derived from the correctness labels, so that both-correct and both-wrong pairs also inform the length and kinship coefficients) on the probability that Assistant A wins, providing a multivariate check on the univariate rates.

3.7. Zero-Cost Constraint

The entire study runs on free API quotas with hard spending protection enabled: every open-source model is capped at its complimentary one-million-token allocation, after which the provider rejects further calls instead of billing them, and the two proprietary judges run on their provider's zero-priced promotional flash tier under platform rate limits. Token consumption was logged per model throughout, and the total footprint of the experiment, 8.4 million tokens across roughly 11,700 API calls, stayed within the complimentary budget. This constraint is not merely logistical; it demonstrates that a full multi-family judge audit is reproducible by any research group at zero monetary cost, which we view as part of the contribution.

4. Results

Two free-quota exhaustions shape the coverage picture and are reported transparently. GLM-5.3 exhausted its complimentary allocation after 714

of 800 main-pass judgements (89%; its coverage is complete for mathematics and nearly complete for knowledge, the two verifiable domains), and Qwen3.8-2.4T completed 782 of 800 (98%); both models could not participate in the test-retest subset, which therefore covers eight open judges. All remaining judges have complete coverage. Agreement statistics use pairwise-available pairs, so partial coverage degrades precision rather than introducing selection bias into the headline metrics. In total the study collected 9,453 usable main-pass verdicts and 640 test-retest verdicts. The proprietary judges completed the main pass in full, but their retest pass was curtailed by platform rate limits and is excluded from headline test-retest statistics; the raw partial records are released with the artifact for transparency.

4.1. Compliance and Decision Severity

Instruction compliance is nearly universal: eleven of twelve judges parse successfully on at least 99% of calls, and the only notable exception is GLM-5.3 at 94.4%, whose mandatory reasoning phase occasionally consumes the reply budget before the verdict token is emitted; the one-reminder parse retry is invoked on only 1.2% of calls overall, rising to 9.4% for GLM-5.3. Compliance, however, is the weakest notion of reliability. Decision severity—how often a judge commits to a winner at all—separates the panel far more sharply, and it separates by family. The five Qwen judges tie on 15–31% of canonical-order pairs, whereas DeepSeek-V4-Flash and DeepSeek-V4-Pro tie on 50% and 59% respectively, GLM-5.3 on 47%, and Kimi-K3 on 44%. The two proprietary references bracket the open panel—Agnes-3.0-Flash ties on 28% of pairs, Agnes-2.5-Flash on 44%—so severity collapse is not an artifact of open-source models or of free tiers specifically; it is a serving-and-tuning

choice that happens to differ systematically across providers. Because ties are excluded from conditional bias statistics and scored as errors in accuracy, this single parameter reshapes every downstream metric for the affected judges, as the following subsections show.

4.2. Ground-Truth Accuracy

Figure 2 reports accuracy on the 66 ground-truth-clear pairs. The Qwen family forms the top cluster (0.64–0.71), with the two flash checkpoints ahead of the larger tiers; GLM-5.3 reaches 0.61 and Kimi-K3 0.50. The DeepSeek family splits: DeepSeek-V4.1-Flash achieves a usable 0.56, but DeepSeek-V4-Flash (0.42) and DeepSeek-V4-Pro (0.32) fall below the 0.5 chance line. DeepSeek-V4-Pro's below-chance score is a direct consequence of severity collapse: on 59% of pairs it abstains with a tie, which the strict scoring counts as an error. Its 0.32 is thus best read as "right one time in three, silent most of the rest" rather than as active misjudgement, and the same reading applies with less force to DeepSeek-V4-Flash. Within-family scale gradients are accordingly non-monotonic for DeepSeek (the flagship Pro tier is the *worst* family member) and roughly flat for Qwen, so capability tier alone is a poor predictor of judging usefulness—a finding we return to in Section 4.9.

4.3. Position Bias and Order Consistency

Four open judges show a statistically significant first-position preference (bootstrap CI excluding 0.5): Qwen3.8-27B at +0.113 and DeepSeek-V4.1-Flash at +0.146 are the strongest, with the two flash Qwens just past the significance line (+0.065 to +0.075). The remaining judges sit near zero,

including the two largest Qwen tiers (-0.013 and -0.028). Average position bias is therefore mild by the standards of early GPT-4-class reports, but the order-swap flip rates expose substantial hidden instability: between 12.5% and 32% of pairs receive different canonical verdicts when the presentation order is swapped. Two judges served with mandatory reasoning are the most order-stable (Qwen3.8-2.4T at 0.125, GLM-5.3 at 0.138), and within the Qwen family the flip rate falls with tier (0.32 and 0.26 for the two flash generations, 0.20–0.31 for the mid and max tiers, 0.13 for the 2.4T flagship). Scale and thinking mode thus buy consistency rather than calibration.

4.4. Length Bias and the Anti-Verbosity Instruction

The judging rubric explicitly instructs models not to reward length. Eight of ten open judges respond by tilting *against* the longer answer (length bias -0.03 to -0.12, significant in the pooled regression: -0.46 log-odds per 1,000 characters of length difference, $p < 10^{-17}$), inverting the verbosity bias routinely reported for rubrics without such an instruction. DeepSeek-V4-Flash sits just on the positive side of zero ($+0.05$), and DeepSeek-V4-Pro at $+0.112$ is the only meaningfully length-favouring judge. The proprietary references show the same anti-length signature, one of them in amplified form (-0.06 and -0.15). In other words, the standard anti-verbosity sentence works—and then some: a one-line instruction meant to neutralize verbosity bias produces a measurable bias in the opposite direction. We attribute the inversion to the instruction itself, since the rubric is otherwise held fixed across all twelve judges, although a no-instruction ablation was not run and the mechanism is therefore inferred rather than demonstrated by counterfactual. This is a concrete, actionable finding for practitioners: the instruction should either be

softened or paired with length standardisation of candidates, and any judge deployed with it should be monitored for short-answer over-preference.

4.5. Family Preference under Ground-Truth Control

Figure 4 and Table 2 summarise kinship behaviour, and the headline is heterogeneity: kinship effects are family-specific rather than universal. Pooling judges within family, Qwen judges pick a same-family answer in only 37.1% of non-tie kin pairs (CI $[0.31, 0.43]$, $n = 950$)—a significant kin-*avoidance*—while GLM-5.3 picks its own family's answers in 63.0% (CI $[0.50, 0.75]$, $n = 81$) and Kimi-K3 and the DeepSeek judges sit near chance (45% and 52%). The quality-controlled statistic sharpens the contrast. When the reference answer proves the kin side wrong, Qwen judges still choose it in only 5.2% of cases ($n = 96$), but Kimi-K3 does so in 33.3% ($n = 9$) and GLM-5.3 in 27.3% ($n = 11$); the GLM and Kimi kin-wrong samples are small, so we read them as directional evidence rather than precise estimates. GLM-5.3 is the sharpest illustration of preference overriding evidence: on the eleven ground-truth-clear kin pairs where its own family is right it picks kin on every one (95% CI $[0.72, 1.00]$), yet on the eleven where its family is wrong it still picks kin more than one time in four. The pooled logistic regression (Section 3.6) returns a kinship coefficient of -0.289 ($p = 2 \times 10^{-4}$), which aggregates the dominant Qwen avoidance with the GLM/Kimi preference; we emphasise that the pooled sign is a property of this roster's family mix, not a law of self-preference. These results generalise self-preference from the individual model to the model *family*, and show that its direction is an empirical, per-lineage question that a ten-minute ground-truth probe can answer before a judge is trusted with high-stakes evaluation.

4.6. Agreement Structure

Pairwise Cohen's kappa on canonical-order verdicts spans 0.27–0.72 with a median of 0.54 (Figure 5)—moderate agreement, broadly in line with published audits of commercial judges. Three structural patterns stand out. First, the highest-agreement pairs sit within the Qwen family: the two Qwen3.8 flash generations agree at $\kappa = 0.72$, and the max and 2.4T tiers at 0.70; the strongest cross-family pairs reach the same level (Qwen3.8-2.4T with GLM-5.3 at 0.70, GLM-5.3 with Kimi-K3 at 0.70). The open judges agree with the proprietary references (mean $\kappa = 0.53$) slightly less than they agree with each other (0.54), so proprietary status is neither a guarantee of a distinct viewpoint nor a privileged one. Second, the family-blocked means reveal a cross-family fault line between Qwen and DeepSeek ($\kappa = 0.44$, the lowest open–open block) while GLM pairs strongly with Kimi (0.70) and with Agnes-2.5-Flash (0.64). Third, the judges whose severity collapsed are also the least agreeable ones: DeepSeek-V4-Pro's mean kappa of 0.43 and the minimum pair Qwen3.7-Flash versus DeepSeek-V4-Pro at 0.27 show that abstention behaviour, not disagreement about winners, is what most damages raw agreement statistics.

4.7. Test–Retest Stability

Repeating a 40-pair subset in canonical order at temperature zero yields agreement between repeats of 0.75–0.98 across the eight open judges with remaining quota. DeepSeek-V4-Pro is the least self-consistent (0.75) and Qwen3.8-27B the most (0.98); the Qwen flash checkpoints sit at 0.85. Stability tracks neither family nor tier cleanly, and notably it does not rescue the weak judges: DeepSeek-V4-Pro is reasonably stable in *abstaining*, which

inflates raw consistency while contributing nothing to discrimination. This decoupling of consistency from validity echoes recent multi-judge audits and cautions against using test–retest agreement alone as a certification of judge quality.

4.8. Panel Behaviour: Accuracy Gains without Debiasing

Figure 7 evaluates majority-vote panels of k judges drawn uniformly from the ten open judges. A panel of three lifts ground-truth accuracy from a single-judge mean of 0.58 (best single 0.71) to 0.85 (CI $[0.80, 0.89]$)—above even the best order-swapped single judge (0.70), which order swapping reaches only by converting 21–67% of pairs into effective abstentions; enlarging to five adds nothing further (0.85), and panels drawn from all twelve judges including the proprietary references reach the same level (0.85), a negligible increment at twice the coordination cost. Position bias, however, is neither amplified nor cancelled: on the same ground-truth-clear pair set, mean first-position preference is +0.04 for single judges, +0.04 for panels of three (CI $[0.00, 0.07]$), and +0.04 for panels of five. The mechanism is straightforward: only a minority of judges carry a significant position lean, and the two-of-three majority requirement outvotes isolated leans while the shared component passes through essentially unchanged. Majority panels are thus an accuracy instrument, not a debiasing instrument, and must still be combined with order swapping or position-balanced aggregation.

4.9. Scale Trends within Families

Figure 6 plots accuracy, absolute position bias, and flip rate against within-family tier. The Qwen family shows a clean and, to our knowledge,

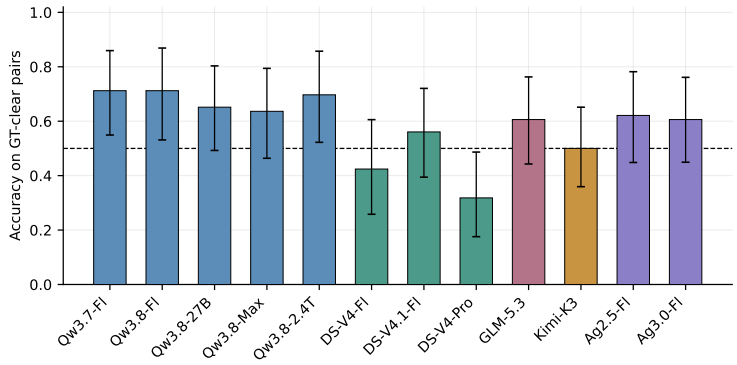


Figure 2: Ground-truth accuracy on the 66 ground-truth-clear pairs (canonical order), with question-level cluster bootstrap 95% CIs. The dashed line marks chance. The proprietary Agnes references are shown in purple.

previously unreported gradient in *consistency*: flip rate falls from 0.26–0.32 across the flash and mid tiers to 0.13 for the 2.4T flagship, while accuracy stays within a narrow 0.64–0.71 band. The DeepSeek family inverts the intuitive ordering: the Pro flagship is the least accurate and least agreeable family member because of severity collapse. Family identity and tuning choices dominate raw parameter count as determinants of judge behaviour, which is why the practical guidance of the next section keys on screening rather than on scale.

5. Discussion and Practical Guidelines

The audit yields one message above all: *free judges are usable, but only after screening, and never naively.* The properties that matter most are not the ones model cards advertise. Parameter count and flagship status predict neither accuracy nor severity; the DeepSeek Pro flagship is the least useful judge in the roster precisely because it abstains on most pairs, while a small

flash checkpoint from the same provider discriminates better. What predicts behaviour is a combination of provider-level tuning (severity), instruction sensitivity (length), lineage (kinship), and deployment configuration (thinking mode for order stability). We distill the findings into six guidelines for zero-cost judging.

G1: Screen for severity collapse first.. Before deploying any free judge, run a twenty-pair probe with a known quality difference and measure the tie rate. A judge that ties on more than about 40% of clearly-different pairs (as the DeepSeek checkpoints, GLM-5.3, Kimi-K3, and one of the two proprietary references do here) will undermine any downstream use that assumes discrimination—leaderboards, win-rate estimation, best-of-n selection—regardless of how sound its individual verdicts are. Severity is the cheapest possible screen: it requires no ground truth, only variation.

G2: Prefer Qwen mid and flagship tiers as workhorses.. Within this roster, Qwen3.8-27B, Qwen3.8-Max, and Qwen3.8-2.4T offer the best combination of accuracy (0.64–0.70), moderate severity (tie rates 0.24–0.31), and order stability (flip rates 0.13–0.31). The 2.4T flagship, served with mandatory reasoning, is the single most order-stable judge and shows no measurable position bias, at the cost of roughly double the token consumption of the flash tiers.

G3: Keep order swapping; add panel position balancing.. Even the most order-stable judge flips on one pair in eight, and the least stable on one in three. Order swapping remains mandatory. Our panel results add a subtler warning: majority voting over judges passes the *shared* position component

through essentially unchanged (mean bias +0.04 for panels versus +0.04 for singles on the same pair set), so panels should aggregate over order-swapped judgements—for example by judging each pair in both orders and requiring the majority to agree on the canonical verdict—rather than over raw single-order verdicts.

G4: Soften the anti-verbosity instruction.. The standard "do not reward length" sentence overshoots into measurable short-answer preference (-0.46 log-odds per 1,000 characters). Either soften it to "length alone does not determine quality", or standardise candidate lengths before comparison, and re-check length bias after any prompt change with a length-stratified probe.

G5: Audit kinship with ground truth before trusting a judge on its own ecosystem.. Family preference is real but lineage-specific: the Qwen judges significantly avoid their own family's answers, while GLM-5.3 and Kimi-K3 reward theirs even when the reference proves the kin answer wrong (27–33% kin-wrong pick rate on small samples). Any evaluation that uses a judge from family X to rank models that include family X should run a ground-truth-controlled kin probe first; the entire probe costs a few hundred free-tier tokens.

G6: Build panels across families, not across tiers.. Agreement structure shows family-level fault lines (Qwen–DeepSeek mean $\kappa = 0.44$), so a three-judge panel drawn from three families decorrelates errors far more effectively than three tiers of the same family. A panel of three open judges reaches 0.85 accuracy on ground-truth-clear pairs—better than the best single free judge (0.71) and statistically indistinguishable from panels that include proprietary

references (0.85)—at zero monetary cost.

Relation to prior work.. Our findings refine three strands of the literature. The verbosity-bias literature documents that judges favour longer answers; we show that the standard mitigation instruction flips the sign of the effect, a prompt-sensitivity phenomenon that length-controlled win rates do not capture. The self-preference literature establishes that judges recognise and reward their own generations; we generalise the unit of analysis from model to family and show that the effect's *direction* varies by lineage, which matters for the increasingly common situation where one provider both ships the judge and supplies some of the candidates. Finally, panel proposals argue that judge diversity averages away idiosyncratic errors; our panel analysis refines this: diversity cancels the independent noise component strongly enough to lift accuracy markedly, but the shared position component passes through majority voting unchanged, so panels must still pair diversity with order-balanced aggregation.

6. Limitations

Our findings should be read against several limitations. First, the ten open-source judges are all served through a single provider endpoint, and the two proprietary judges through a second one. Although this removes serving differences as a confound within each group, it also means that our observations reflect the providers' quantization and serving configurations of each checkpoint rather than the raw open weights, and replication on self-hosted inference may differ in detail; in particular, the Agnes platform injects undisclosed system instructions ahead of our judging prompt, so the

proprietary judges' absolute numbers are indicative rather than strictly comparable to the open panel. Second, the roster covers four model families that all originate from Chinese development teams, because these are the families with usable free tiers on the chosen platform at the time of the study; Western open families such as Llama or Mistral were not available under the zero-cost constraint, so our family-preference findings should be generalized to those families with care, and the GLM family is represented by a single checkpoint, so GLM-specific conclusions rest on one observation. Third, two judges are served with a mandatory reasoning phase, which is entangled with their family identity and tier; we treat thinking mode as an observational variable and report where it plausibly matters, but a fully crossed thinking-by-family design is not realizable on this platform. Fourth, the evaluation uses 400 pairs over four English-language domains. While the stratified design gives every judge the same stimulus set and the bootstrap accounts for question-level clustering, the absolute number of ground-truth-clear kin pairs per family remains modest, and kin-preference estimates for individual judges carry correspondingly wide intervals. Fifth, our generators produce single answers per question at a moderate sampling temperature; verdicts could vary with additional samples, and we leave multi-sample generation to future work. Finally, the two open-ended domains lack ground truth, so for those domains our reliability claims are about agreement among judges rather than closeness to correctness.

7. Conclusion

We audited the judging behaviour of every LLM reachable at zero monetary cost on two public platforms at the time of study: ten open-source judges from four families and two proprietary free-tier references, evaluated under one protocol over 400 answer pairs in four domains, with both presentation orders, a test–retest subset, and ground-truth-controlled family-preference probes. The picture that emerges is nuanced but actionable. Free judges are viable: the strongest open checkpoints reach 0.71 accuracy on verifiable pairs and inter-judge agreement comparable to published commercial audits, and a three-judge cross-family panel of free models beats the best single free judge. But viability is conditional on screening, because the failure modes are systematic rather than random: severity collapse that silences entire families, an anti-verbosity instruction that overshoots into short-answer preference, lineage-specific kin favouritism—within the four families studied—that survives ground-truth control, and majority voting that inherits, rather than removes, the shared position bias. Every one of these failure modes is detectable with probes that cost a few hundred tokens, and every probe ships with the released artifact, so the audit and its screening kit can be re-run by any group as free-tier catalogues evolve. As evaluation infrastructure comes to rest on machine judgement, the reliability of the judges themselves—especially the ones that cost nothing—deserves the same systematic attention that has been lavished on the systems they grade.

Data and Code Availability

All prompts, the 200-question evaluation set, raw verdict logs, and analysis scripts are released at `github.com/43aquarius/LLM-as-a-Judge`.

Acknowledgements

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Table 1: Model roster. Judges evaluate all 400 answer pairs in both orders; generators produce the candidate answers. Tier gives the capability level within the family (F = flash/entry, M = mid, P = flagship/prime). Thinking indicates whether the serving platform enables a reasoning phase that cannot be disabled. Agnes judges are proprietary free-tier references (see text).

Model	Family	Tier	Role	Thinking
Qwen3.7-Flash (0715)	Qwen	F	judge	no
Qwen3.8-Flash	Qwen	F	judge + generator	no
Qwen3.8-27B	Qwen	M	judge	no
Qwen3.8-Max (0902)	Qwen	P	judge + generator	no
Qwen3.8-2.4T-A95B	Qwen	P+	judge	yes
DeepSeek-V4-Flash (0731)	DeepSeek	F	judge	no
DeepSeek-V4.1-Flash	DeepSeek	F	judge	no
DeepSeek-V4-Pro (0813)	DeepSeek	P	judge + generator	no
GLM-5.3	GLM	P	judge + generator	yes
Kimi-K3	Kimi	P	judge + generator	no
Agnes-2.5-Flash	Agnes[p]	F	judge	managed
Agnes-3.0-Flash	Agnes[p]	F+	judge	managed

[p]Proprietary judges, served via their own API.

Table 2: Per-judge summary on the main pass. Tie is the rate of [[C]] verdicts in canonical order; Acc is accuracy on the 66 ground-truth-clear pairs; PB is position bias (first-win rate among non-ties minus 0.5); Flip is the order-swap verdict-flip rate; LB is length bias (longer-answer win rate among non-ties minus 0.5); κ is the mean pairwise Cohen's kappa against the other judges; Retest is test-retest agreement on the repeated subset ("–" where the model exhausted its free quota or platform rate limits curtailed the retest).

Judge	Tie	Acc	PB	Flip	LB	κ	Retest
Qwen3.7-Flash	0.15	0.71	0.075	0.32	−0.072	0.46	0.85
Qwen3.8-Flash	0.29	0.71	0.065	0.26	−0.082	0.60	0.85
Qwen3.8-27B	0.24	0.65	0.113	0.31	−0.083	0.56	0.97
Qwen3.8-Max	0.24	0.64	−0.013	0.20	−0.063	0.57	0.95
Qwen3.8-2.4T	0.31	0.70	−0.028	0.13	−0.099	0.59	–
DeepSeek-V4-Flash	0.50	0.42	0.043	0.29	0.048	0.47	0.90
DeepSeek-V4.1-Flash	0.44	0.56	0.146	0.28	−0.034	0.55	0.82
DeepSeek-V4-Pro	0.59	0.32	0.070	0.25	0.112	0.43	0.75
GLM-5.3	0.47	0.61	0.019	0.14	−0.124	0.60	–
Kimi-K3	0.44	0.50	0.042	0.25	−0.051	0.58	0.90
Agnes-2.5-Flash^p	0.44	0.62	−0.052	0.27	−0.150	0.52	–
Agnes-3.0-Flash^p	0.28	0.61	−0.030	0.33	−0.057	0.54	–

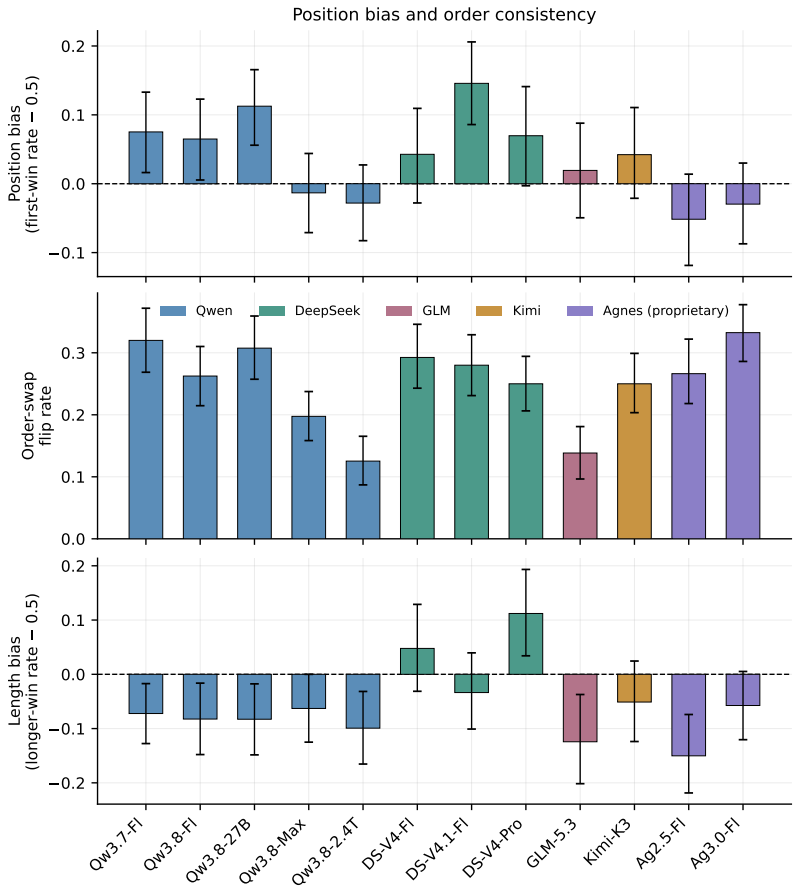


Figure 3: Bias trio per judge with 95% CIs: position bias (top), order-swap flip rate (middle), and length bias (bottom). Dashed lines mark the unbiased null.

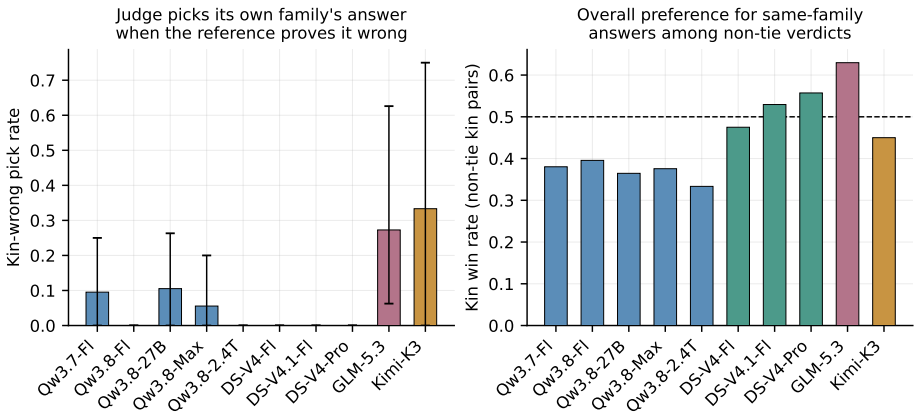


Figure 4: Family preference for the ten open judges. Left: kin-wrong pick rate, the fraction of ground-truth-clear kin pairs in which the judge picks its own family's answer although the reference proves it wrong. Right: overall kin win rate among non-tie kin pairs (dashed line: 0.5).

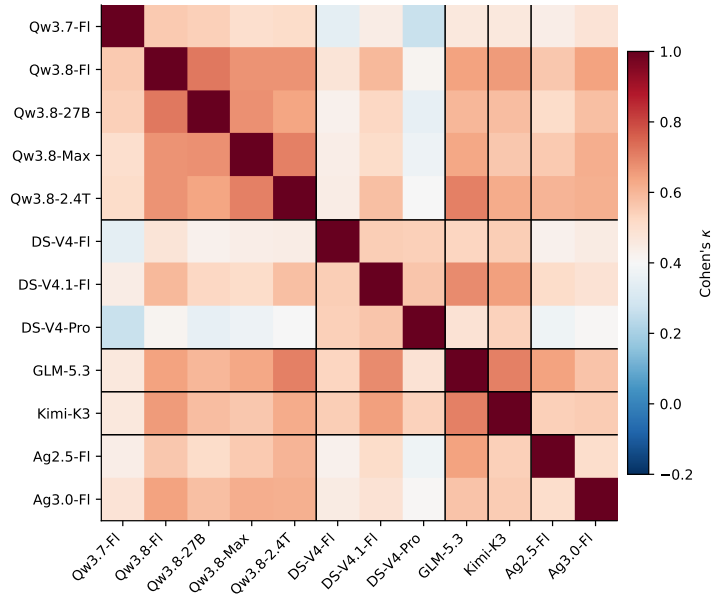


Figure 5: Pairwise Cohen's kappa between judges on canonical-order verdicts. Black grid lines separate model families.

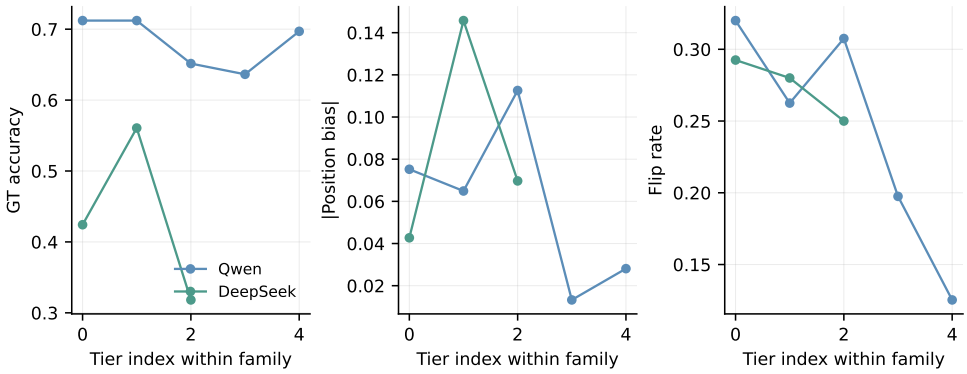


Figure 6: Within-family trends across capability tiers for families with at least two tiers: ground-truth accuracy (left), absolute position bias (middle), and order-swap flip rate (right).

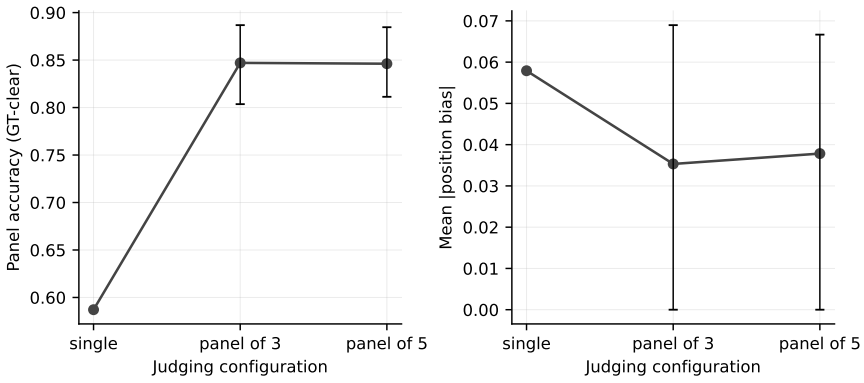


Figure 7: Majority-vote panels of open judges: ground-truth accuracy (left) and mean position bias (right) for single judges and panels of three and five. Panels raise accuracy but pass the shared position component through essentially unchanged.